

To what extent does the River Modau conform to the Bradshaw Model?



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1 Introduction

1.1 Aim

The aim of this following research is to determine the accuracy of the Bradshaw Model in predicting the characteristics of a river as it flows from its source down to its mouth. In modern times, many anthropogenic changes to natural landforms have been made, either to accommodate growing settlements or to make way for various projects at an ideal, specific location. To evaluate the impact of these modifications on, and whether all rivers do act in accordance to, the Bradshaw model, the following question, “*To what extent does the River Modau conform to the Bradshaw Model?*”, was formulated.

The survey for this study was conducted on the 26th of June, 2025 between the hours 8 a.m. to 4 p.m.. This investigation adheres to Option A: Freshwater, specifically **Drainage Basin Hydrology** and **Geomorphology** (Clastify).

1.2 Bradshaw Model

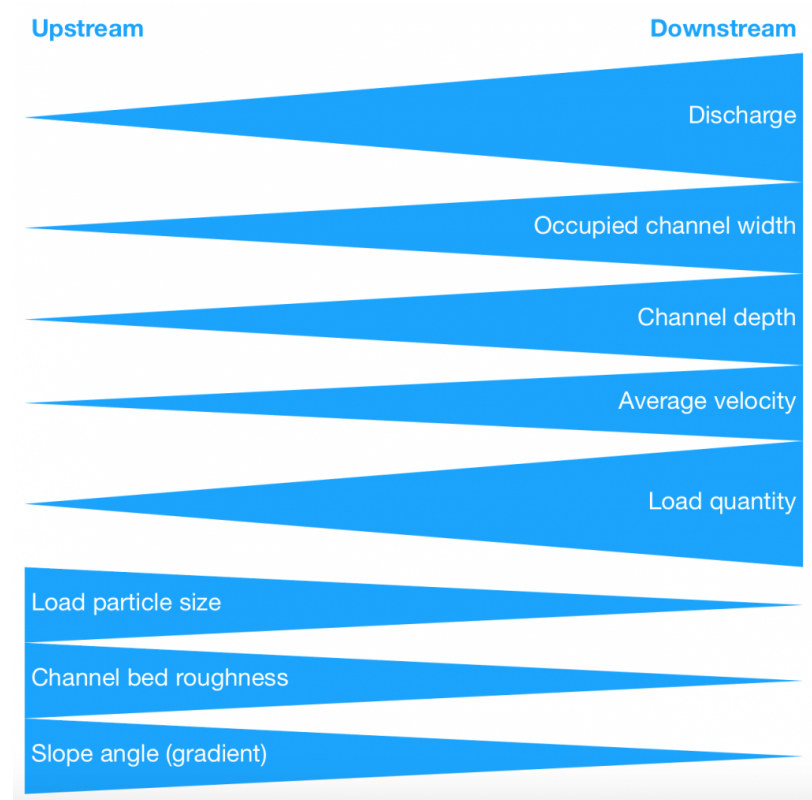


Figure 1: Bradshaw Model (Internet Geography)

The Bradshaw Model is a geographical model which theorizes the characteristics of a river, stream, or tributary as one moves downstream from the source to the mouth (En-Academic). However, the model can be read from both ways, upstream and downstream. Here it portrays how river discharge, channel width, depth, average velocity, and load quantity increase as one proceeds downstream. On the other hand, moving down stream, the load particle size, channel bed roughness, and gradient decrease (Cool Geography). Despite showing the predicted river characteristics moving downstream, the Bradshaw Model does not encompass all the features of the river. Properties such as channel efficiency, river valley shapes and river landforms are exempt from this paradigm.

1.3 Modau River

The Modau River is a small river located in the southern part of the state of Hesse, Germany. Geographically, it originates in the Odenwald hills near Neunkirchen and flows northwest through a series of valleys and lowlands.

The river stretches approximately 44 kilometres before it empties into the Rhine River near Stockstadt am Rhine.

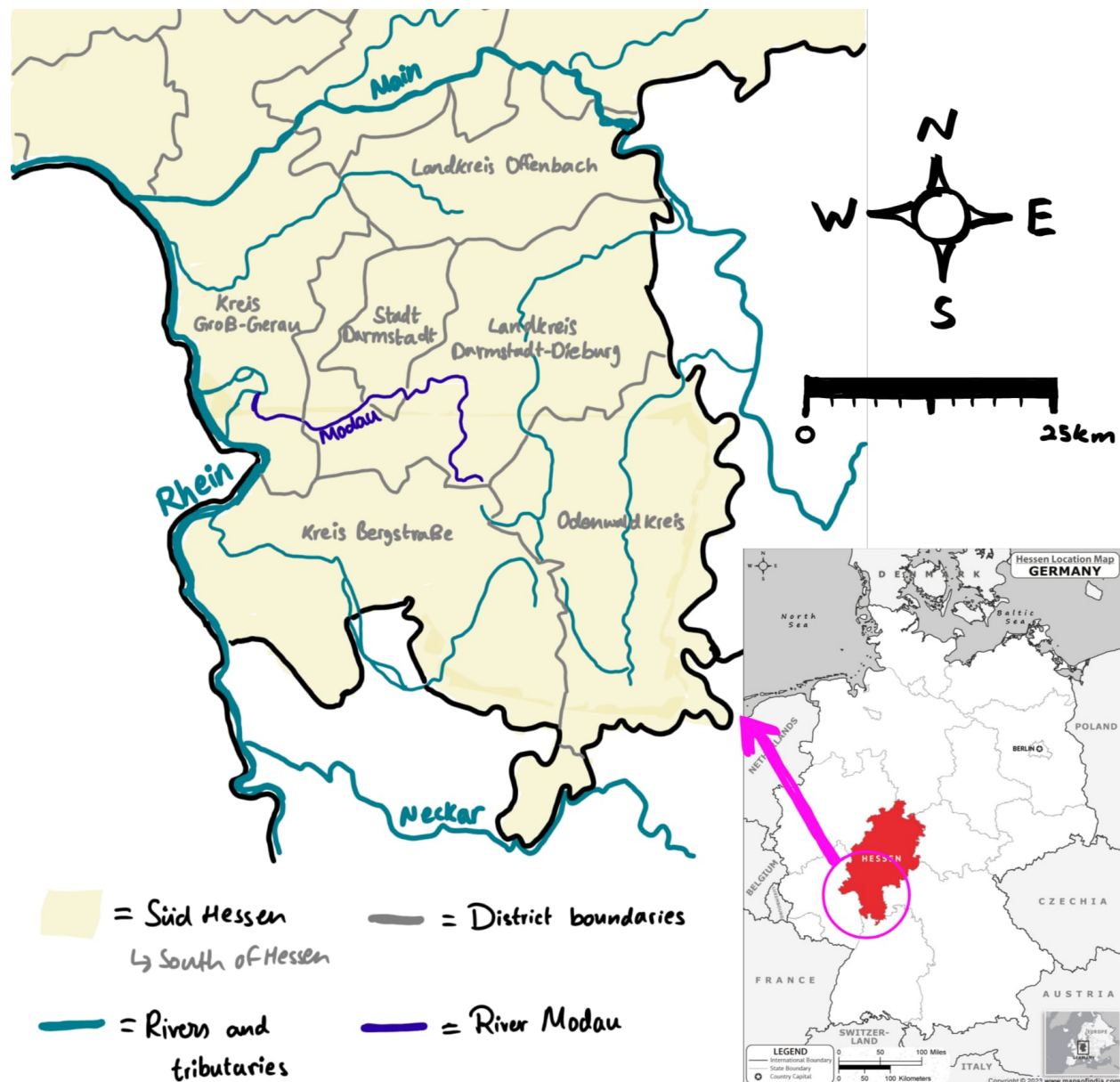


Figure 2: River Modau geographical location in relation to Map of Germany (MapsofIndia)

Its drainage basin lies within a temperate climate zone, and the river plays a role in regional hydrology, especially in terms of groundwater surface-water runoff.

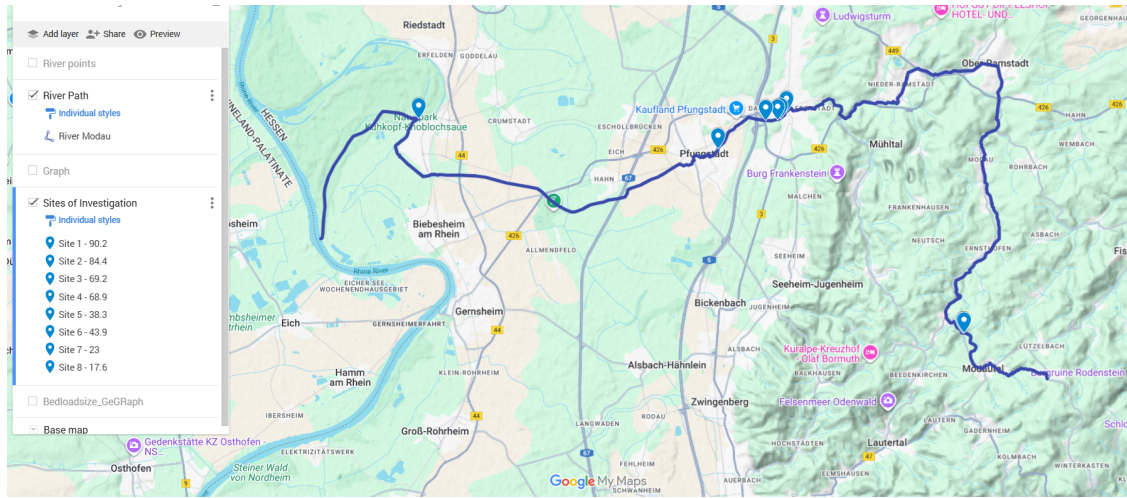


Figure 3: Map of Modau and the sites of investigation created using Google Maps.

The map shows the Modaus full course: starting in the eastern hills with a straight, V-shaped upper valley, moving into a flatter middle course near Pfungstadt with gentle meanders, and finally entering a wide, flat valley before joining the Rhine. The meanders become more visible downstream, and additional tributaries join the Modau as the altitude decreases.

2 Hypotheses

Hypothesis 1: If one moves downstream, then the **velocity**¹ increases as **channel efficiency** increases.

This is as channel efficiency is entirely dependent on the **hydraulic radius**.² An increase in hydraulic radius leads to an increase in channel efficiency, which means the proportion of water in contact with the river banks and bed is less, meaning there is less drag, allowing the velocity to increase.

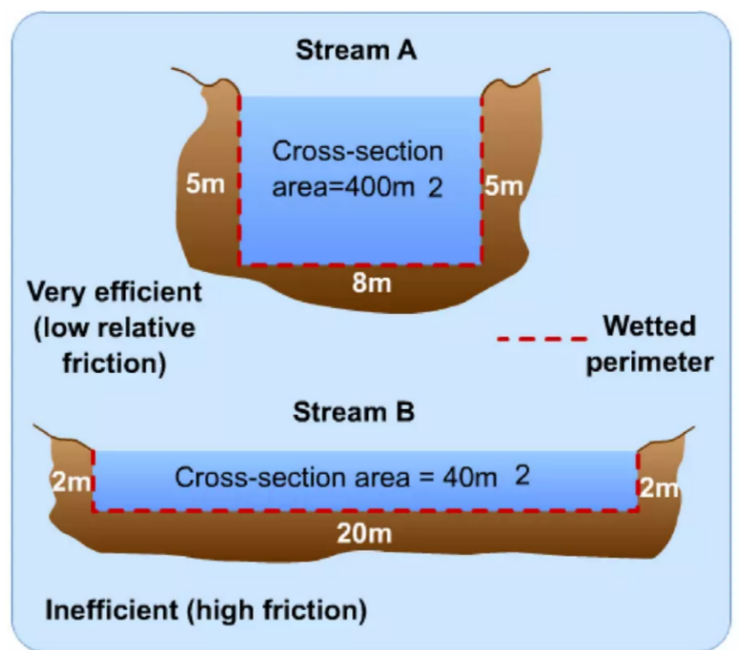


Figure 4: Channel Efficiency example (Slideshare).

As portrayed above, a semi-circular channel shape provides less contact with the river banks and bed compared to a flat floored channel, meaning the semi-circular channel has a higher channel efficiency. Furthermore, there is more vertical and lateral erosion as one moves downstream, increasing channel efficiency as the channel walls get smoother, causing less friction.

¹The speed at which the river flows measured in either m s^{-1} or km s^{-1} .

²The cross-sectional area divided by the wetted perimeter provides the hydraulic radius.

Hypothesis 2: If one moves downstream, then there is an increase in **river discharge**³, because **channel width**⁴ and **channel depth**⁵ increase.

This occurs as the more downstream one goes, the more the water erodes the banks and the river bed through abrasion due to its sediments, increasing channel depth and width. This increases the carrying capacity of the river, allowing for a greater river discharge.

Hypothesis 3: Bedload⁶ **size** will decrease as we go downstream, as there are more erosional processes taking place due to the increased amount of sediment being carried in the river.

This is reasonable as due to an increased amount of load carried by the river as it moves downstream, there are more erosional processes taking place and to a greater extent. These include processes like traction, corrosion, attrition, and more where sediment drags along the river bed eroding itself and the bed of the river, acidic conditions in the water can cause sediments like limestone to erode, small rocks hitting each other due to the river current causing them to wear each other down, and more.

³The volume of water flowing through a channel at a certain point of time, measured in cumecs ($\text{m}^3 \text{s}^{-1}$).

⁴How wide the river is, measured laterally, in meters, at the surface from river bank to river bank where the water touches the land.

⁵How deep the river bed is from the surface of the water, measured at set intervals perpendicular to river flow in order to account for the irregular shape of the river bed, measured in meters.

⁶The sediment transported along the bottom of the river due to the river flow (Fiveable).

3 Methodology

3.1 Velocity (Digital)

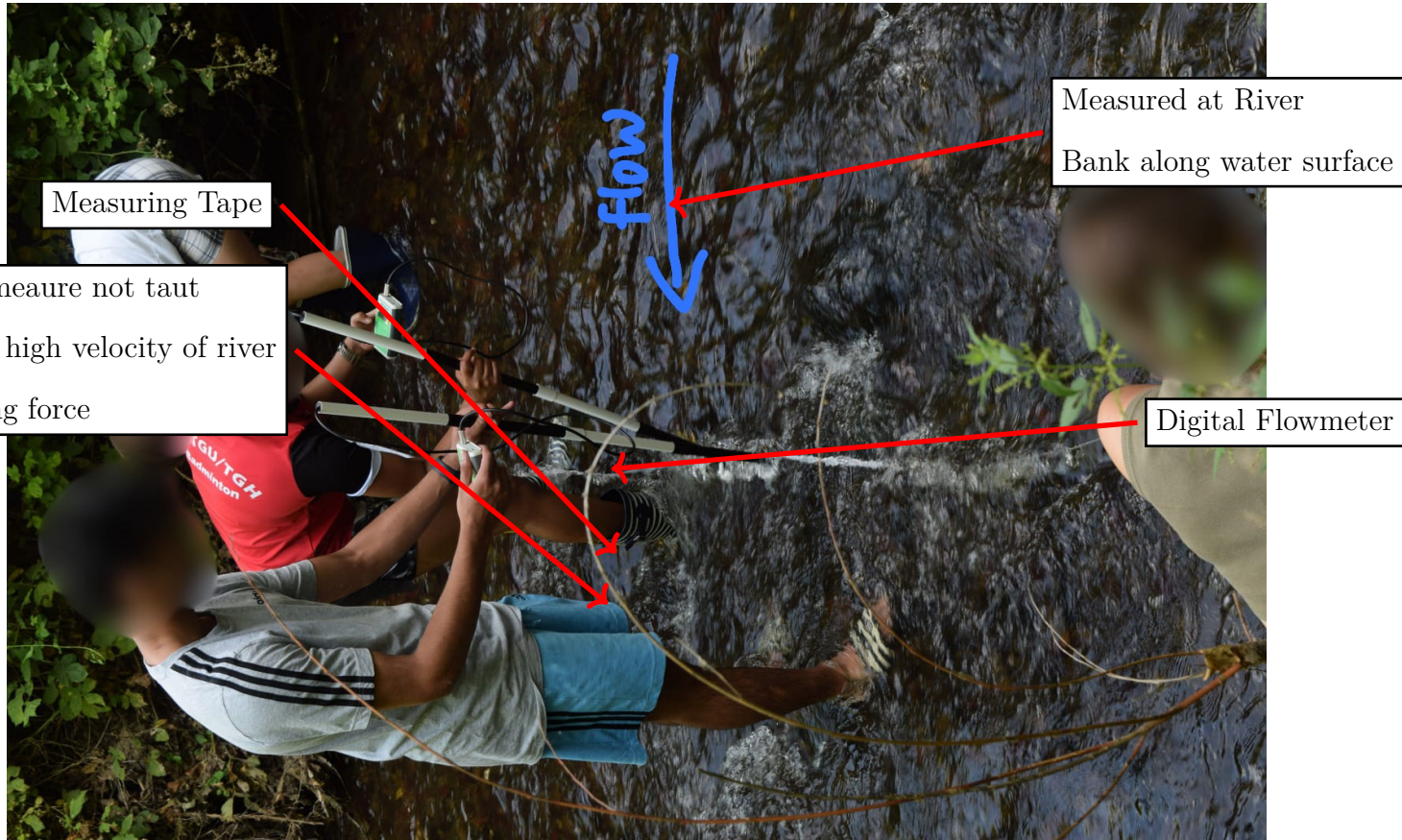


Figure 5: Digital Velocity measured at Site 3.

Equipment

Tape Measure, Ranging pole, Flowmeter and its reader

Method

Velocity was measured in two ways for the following investigation, digital and analogue. To measure digitally, first the width of the river channel was measured at each site and the middle of the channel was recorded. This middle point in the river was where the flow-meter was placed at the surface, with the propeller facing against the flow, in order to measure the

velocity as a form of **purposive sampling**⁷.

Velocity Equation when digital flow-meter is not used:

$$v = \frac{\Delta s}{\Delta t}$$

v = velocity (m/s)
 Δs = change in distance (m)
 Δt = change in time (s)

Justification

Purposive sampling is used, as the least drag is present in the middle of the river channel and away from the river bed, which is a pre-defined characteristic chosen for higher accuracy. This was measured 4 times as the data kept fluctuating by one decimal point, then the average was calculated. A digital flow-meter was used for the data as it is more accurate than an analogue measurements as obstacles in the river, can halt or alter the flow of the float.

The calculation used to find the correlation between data sets was Spearman's ρ as more than 1 data set was used and the relationship between the two were ordinal.

Spearman's Rank Correlation Coefficient

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

ρ = Spearman's rank correlation coefficient

d_i = difference between the paired ranks for observation i

n = number of paired observations

(Formula from: Spearman)

⁷**Purposive sampling** is when a sample is chosen based on a predetermined feature, which is beneficial to the outcome of the study.

3.2 Wetted Perimeter



Figure 6: Measurement of Wetted Perimeter in middle course.

Equipment

Tape measure

Method

The measurement for the wetted perimeter of the channel was done with a normal tape measure neither a weighted tape measure nor a chain was available. The tape measure was pressed down against the river bed and the river banks by the students and was pulled taut for accurate measurements. The measurement start and end points were at the waterline, where the river meets the bank at the surface of the water. The tape measure was then pressed along the side of the river banks till it reached the bed, and then the bed was measured.

Justification

Pragmatic sampling was used here so as to gain proper results areas not obstructed by shrubs or larger bedload were selected.

3.3 Channel Width



Figure 7: Measurement of Channel width in middle course.

Equipment

Tape measure

Method

Measurement of the channel width was taken by using a tape measure. The tape measure was placed perpendicular to river flow and the measurement was taken from river bank to river bank at the points where the water meets the bank. During this the tape measure

was pulled taut across the water surface before the measurements were taken. There were 3 measurements made for width and the sampling technique used was pragmatic sampling.

Justification

This method of sampling was used as the point in the sites chosen to measure width were chosen as they were the most accessible with the least amount of obstructions, which would cause uncertainties.

3.4 Channel Depth

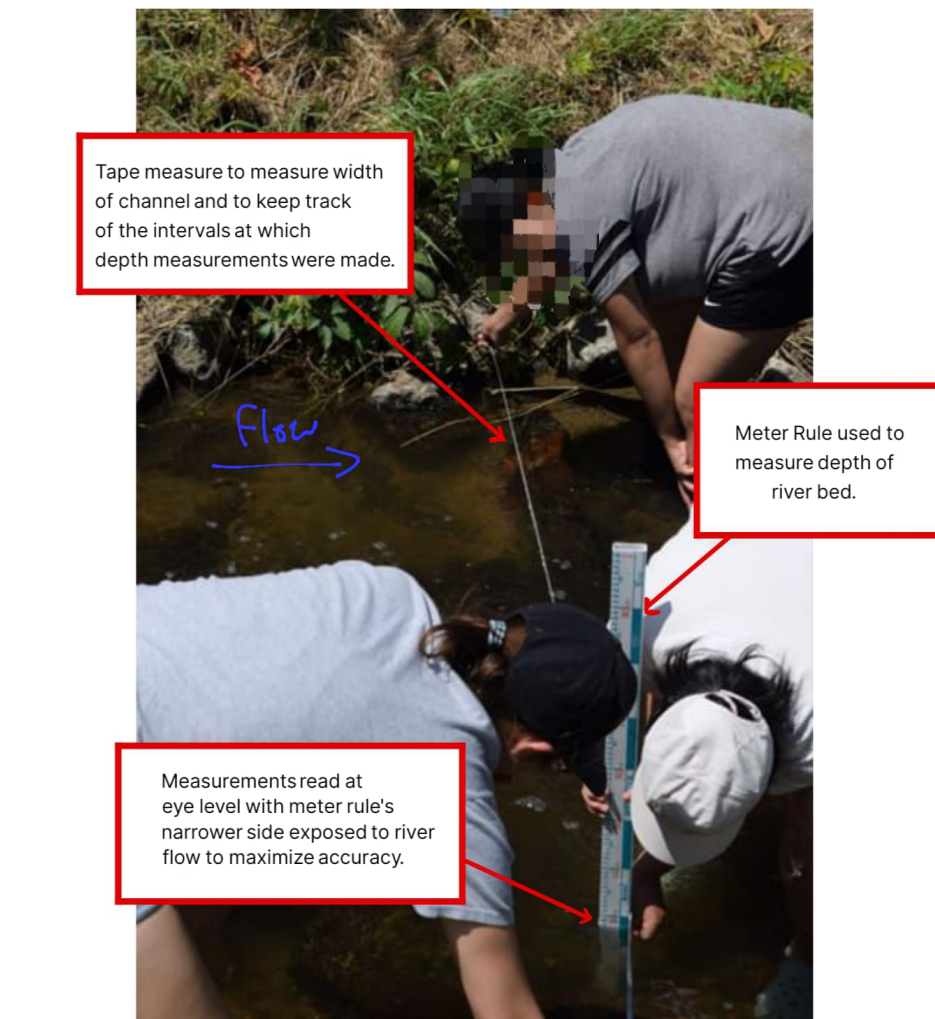


Figure 8: Depth measurements taken at river

Equipment

Tape measure, Meter Rule, Ranging poles

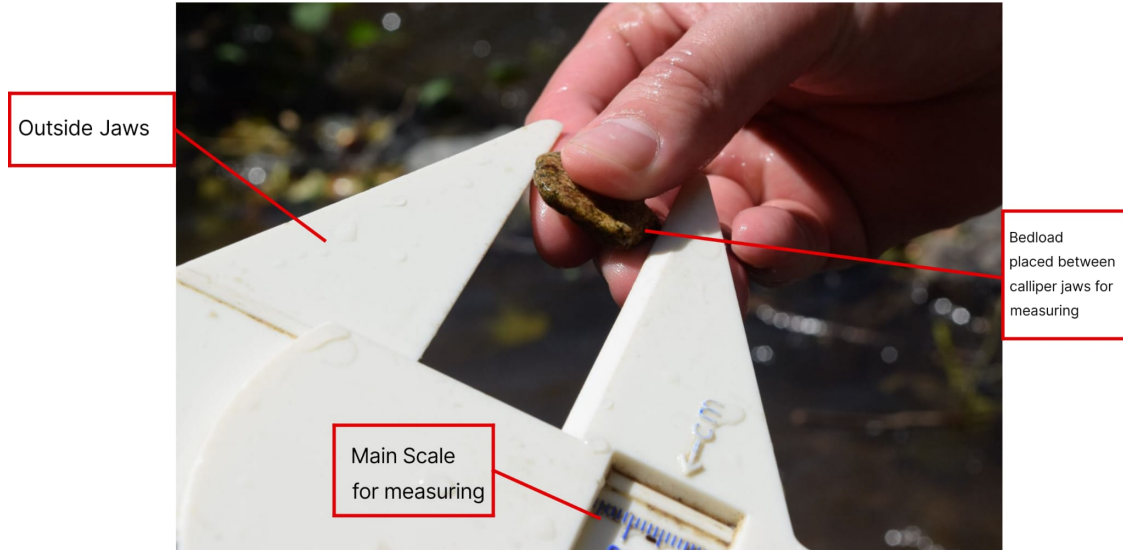
Method

Measurement of the channel depth was taken through the use of a meter rule. A tape measure spanning the width of the river was placed in order to provide the overall width, which would then help determine the intervals at which to measure the depth across the river channel. The meter rule was placed with its narrower side exposed to river flow and the distance between the water surface and the riverbed was measured and recorded.

Justification

Pragmatic sampling was also utilised here as places that had fewer significant protrusions in the river bed, such as large rocks, were avoided to allow for a more accurate measurement of the river bed. These measurements were taken 3 times at each interval and had the average calculated for each of the intervals. Then for the overall depth, the mean of the averages was used.

3.5 Bedload Size



Equipment

Calliper, Bedload samples

Method

To measure bedload size, a calliper was used and measurements on the chosen rock's length, width and breadth were taken. These results were then averaged and recorded on a table. 10 samples were measure at every site, the averages of the average sizes of these samples were taken in order to determine the average bedload size at each site.

Justification

Systematic sampling was used in this case so as to broaden the variety of bedload samples used to be measured in order to provide a higher level of accuracy.

Furthermore the method used to find the correlation between the data sets was Spearman's Rank correlation coefficient:

4 Processed Data

4.1 First Hypothesis

4.1.1 Velocity

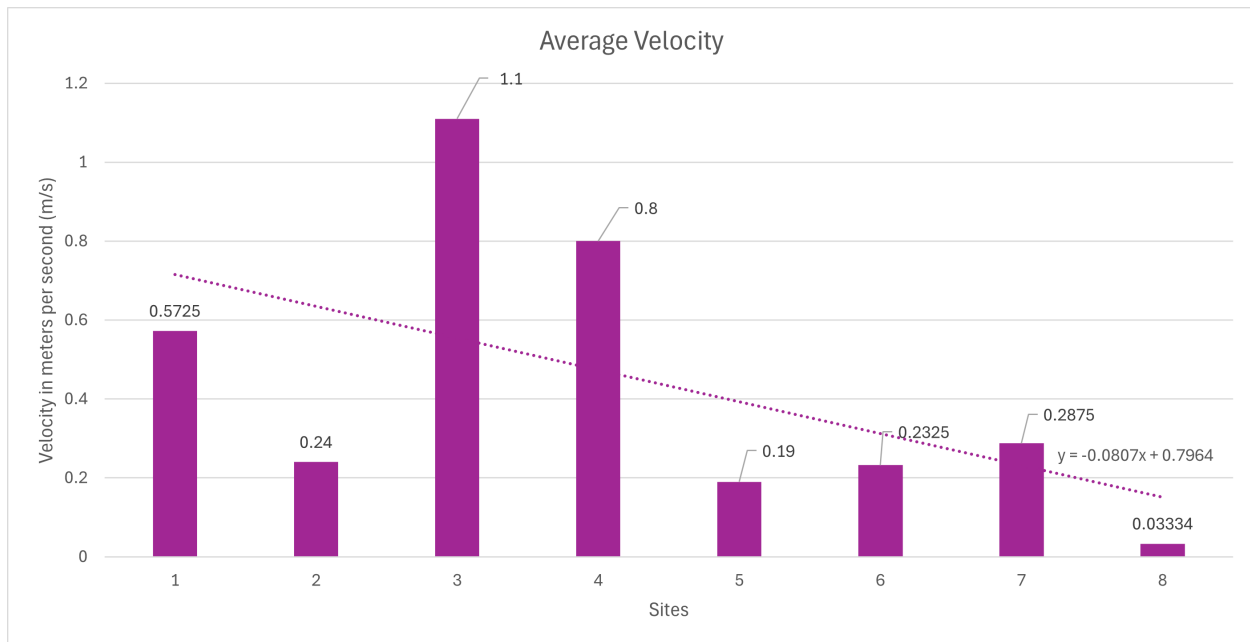


Figure 9: Average Velocity at Each Site

The velocity of a river, according to the Bradshaw model, should increase further downstream due to the predicted widening of the river channel and increase in depth. The river Modau on the other hand, had a decrease at the very start from 0.5725 m/s at Site 1 to 0.24 m/s at Site 2. Site 3 shows a high spike at 1.1 m/s, however following this the velocity decreases again. Only Site 5 to Site 7 embody the characteristics of the Bradshaw Model. This portrays that the velocity of the river does not conform to the Bradshaw Model, shown by the negative value of the line of best fit.

The reason for the very high value at Site 3 is as the channel had been straightened, with matting on the bottom due to the channel's location in a town, with a footpath had been made alongside it.

4.2 Cross-Sectional Area

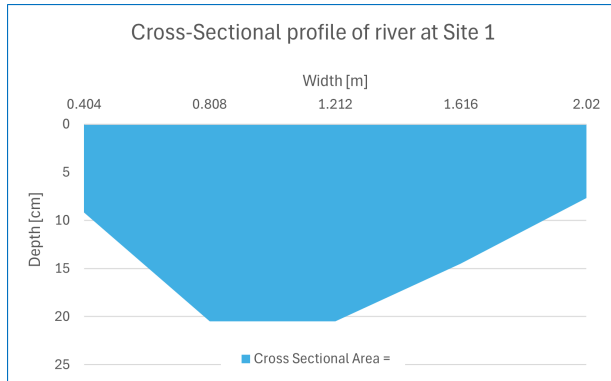


Figure 10: Site 1

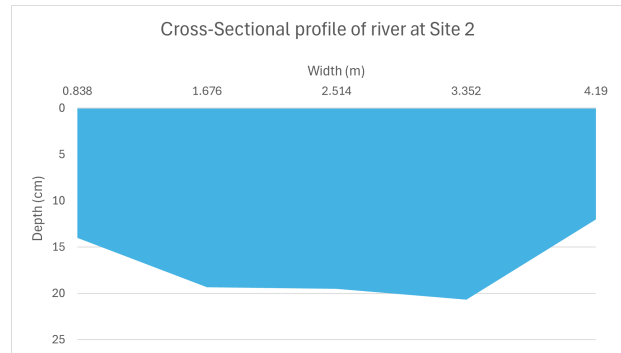


Figure 11: Site 2

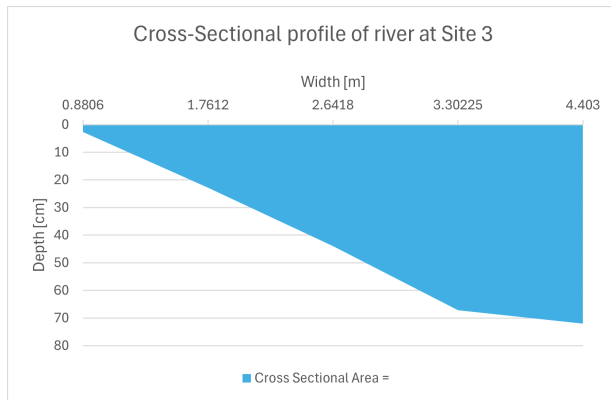


Figure 12: Site 3

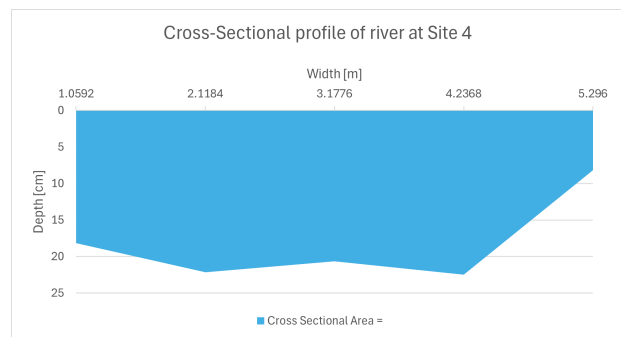


Figure 13: Site 4

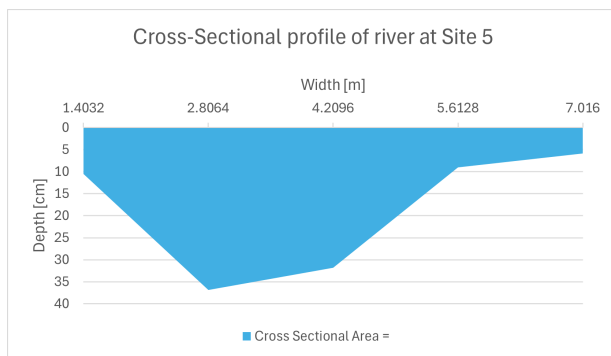


Figure 14: Site 5

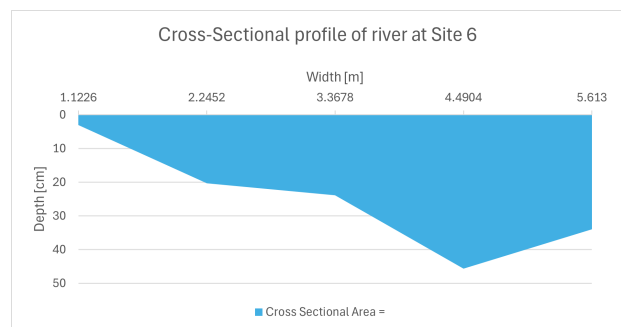


Figure 15: Site 6

In all cross-sectional graphs except for Figure 11 the comma is analogous to a decimal.

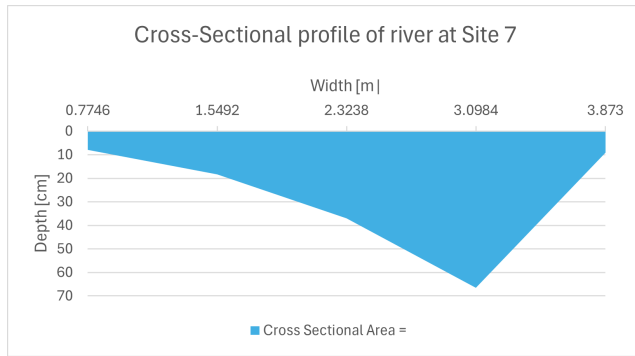


Figure 16: Site 7

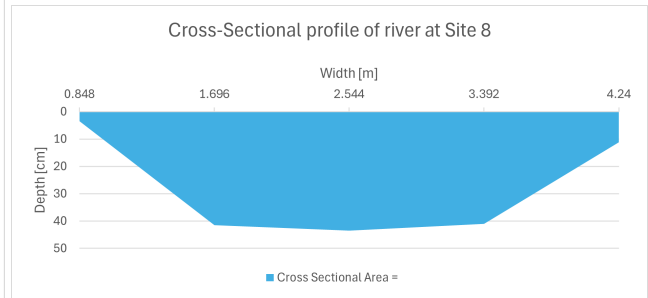


Figure 17: Site 8

The Bradshaw model states that the width and depth of the river channel increases downstream, from which one can infer that the cross-sectional area must also increase downstream. Looking at Figure 10 to Figure 17, one can deduce that the cross-sectional area does increase downstream as implied in the Bradshaw model. One can differ, nonetheless, that some cross-sectional profiles have shallow ends and deeper ends, caused due to the erosional and sedimentational processes occurring at meanders.

4.2.1 Wetted Perimeter

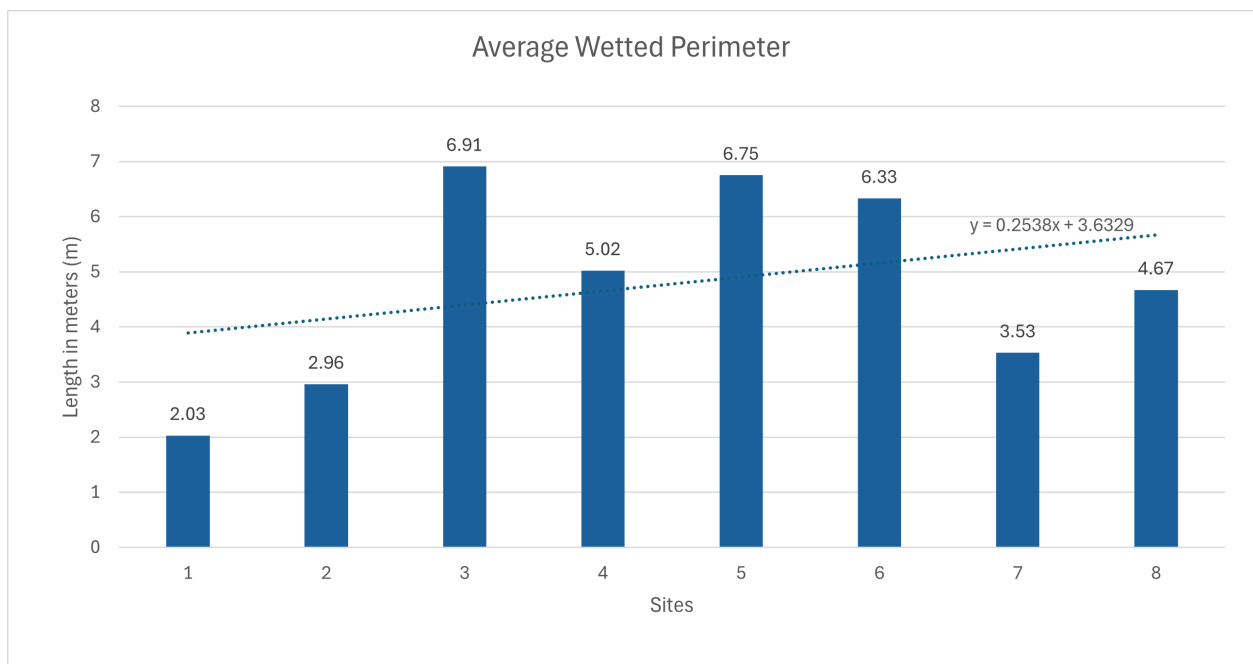


Figure 18: Wetted Perimeter at Different Sites

Wetted Perimeter as well is thought to increase in theory, based on the Bradshaw Model as there is an increase in Depth and Width of the river channel. This is supported by the gathered data as, despite it being low, there is an overall increasing trend in the wetted perimeter as one moves downstream. However there are multiple anomalies as there are sudden drops as seen in Site 4 and Site 7.

4.2.2 Channel Efficiency

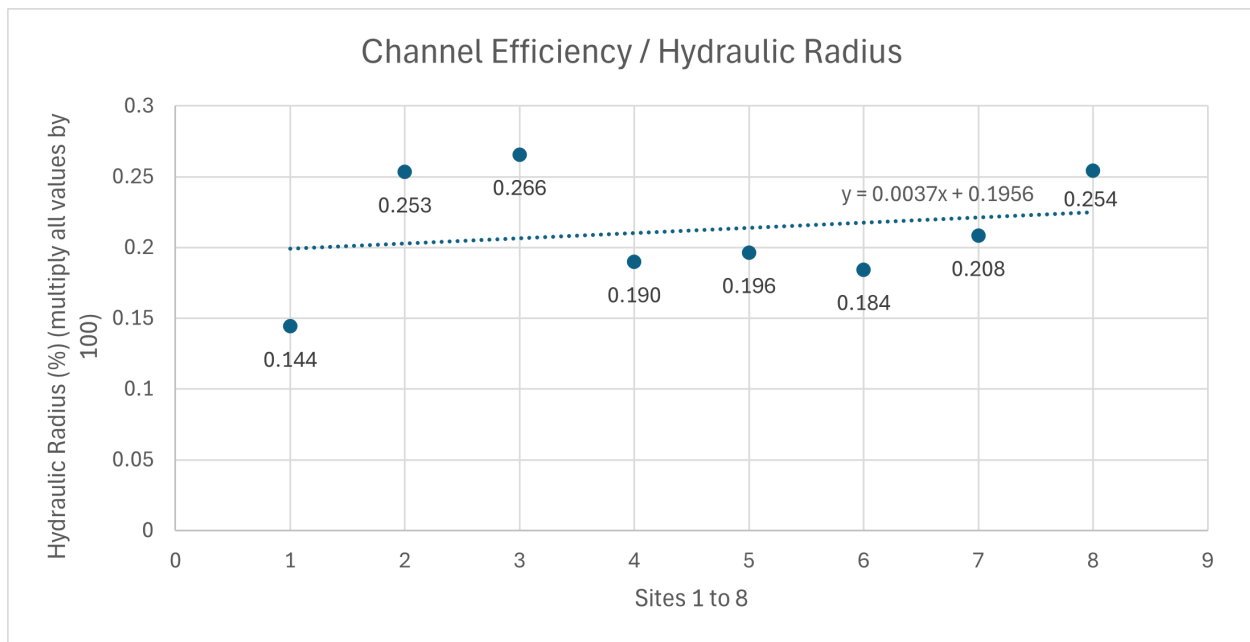


Figure 19: Channel Efficiency at Different Sites

Channel Efficiency, calculated by dividing the cross-sectional area by the wetted perimeter, is said to increase as one moves downstream according to the Bradshaw Model. This is as when depth and width increase, cross-sectional area and wetted perimeter increase. As shown in the graph above channel efficiency fluctuates at the start as there is a sudden, sharp increase from Site 1 at 14.4% to 25.3% at Site 2. The maximum is seen at Site 3 where it reaches a high of 26.6%, however, it drops after this till 19%. From Site 6 to 8, a steady increase is seen. Overall, there is an increasing trend, albeit a very small increase as shown by the line of best fit, but one can see that the channel efficiency corresponds to the

Bradshaw Model.

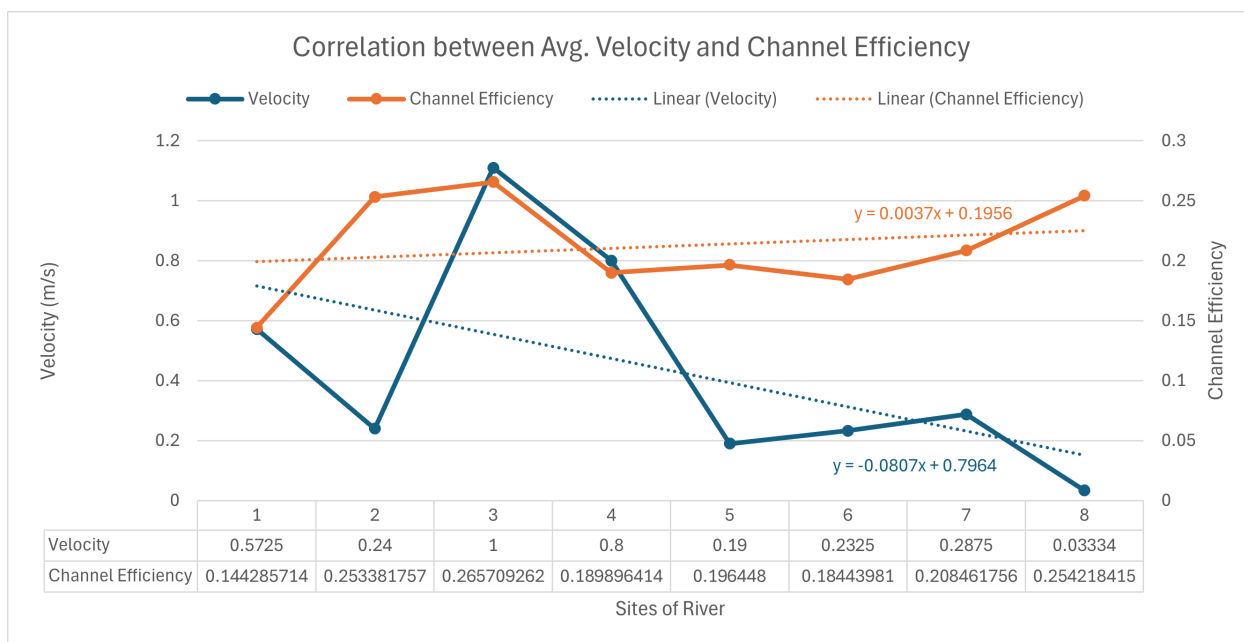


Figure 20: Channel Efficiency and velocity correlation across different sites.

The Bradshaw Model states that both channel efficiency and velocity increase downstream, however, only channel efficiency conforms to the Bradshaw Model. The values of both data sets are mainly similar at sites 2, 5, 6, but this doesn't mean they follow the same trend. Their values only get close to each other here. One can see that despite channel efficiency increasing from site 1 to site 3, velocity fluctuates till Site 3 where it suddenly increases to 1 m/s, its highest speed.

The only similarity is that they both have their maximum values at Site 3, however channel efficiency is at its lowest at Site 1 with 14.4% and velocity is lowest at Site 8 with 0.0333 m/s. This is as Site 8 was the mouth of the river, causing the water to slow because of sedimentation and delta formation. Overall, the datasets show us that the hypothesis partially corresponds to the Bradshaw Model as channel efficiency's overall trend follows the model, however velocity does not. Additionally, the hypothesis is not supported as there is a very weak negative correlation between velocity and channel efficiency with a correlation coefficient (Spearman's Rank) of -0.0238 (3 s.f.).

4.3 Second Hypothesis

4.3.1 Channel Width and Depth

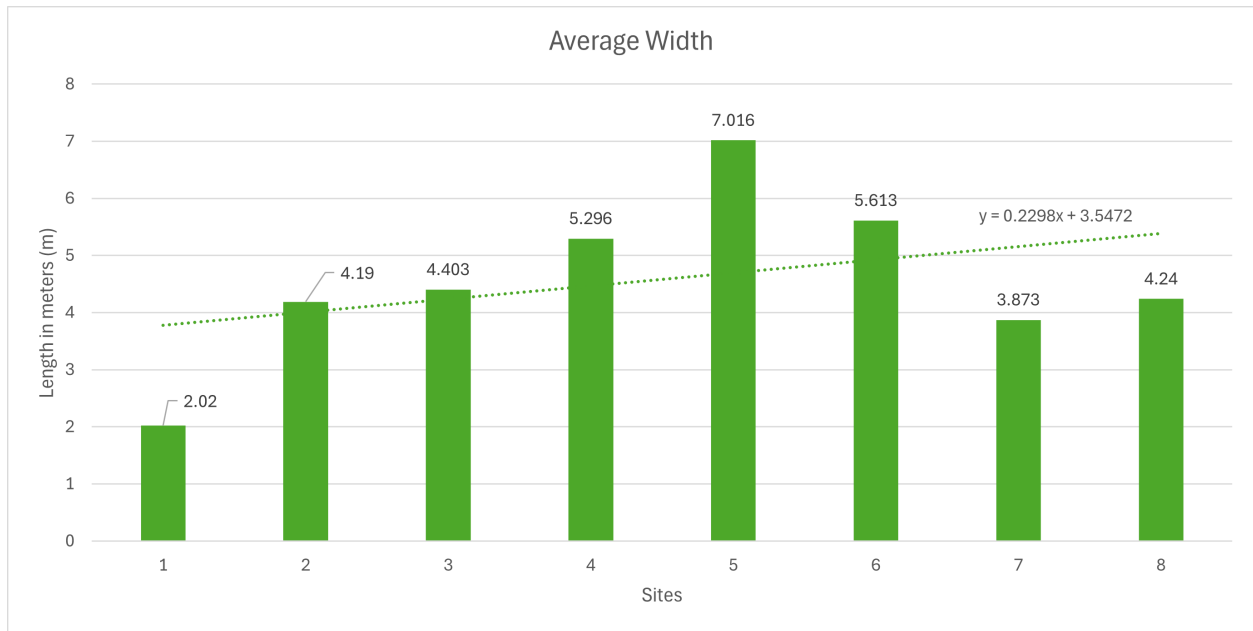


Figure 21: Average Channel Width at Each Site

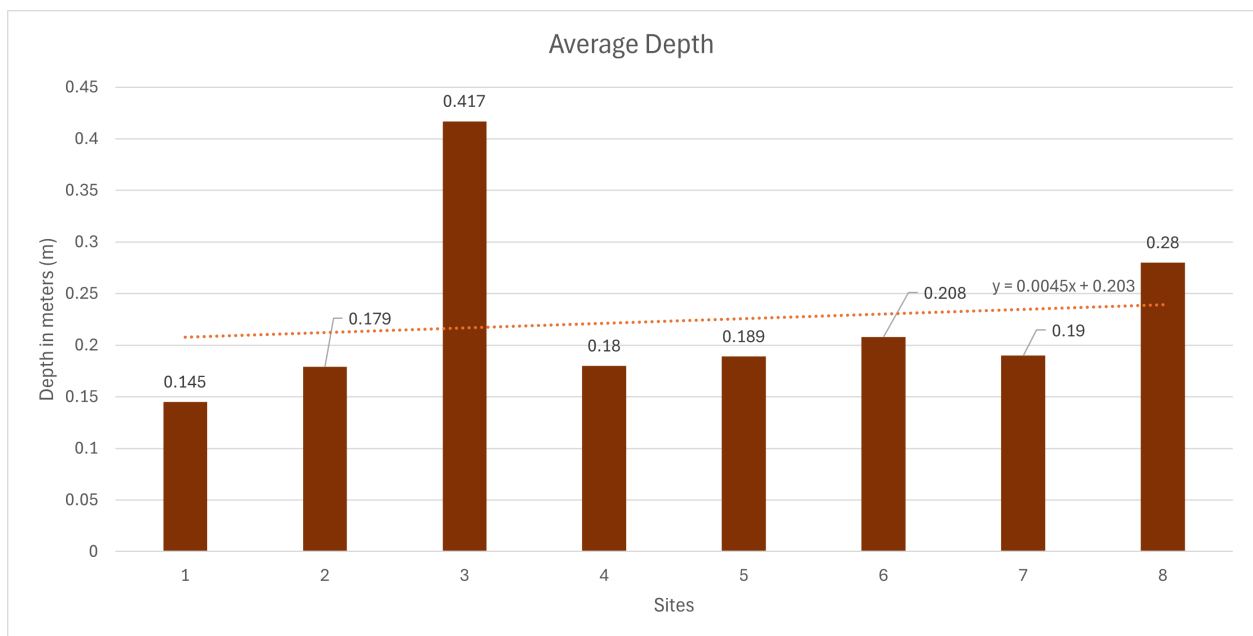


Figure 22: Average Channel Depth at Each Site

The aforementioned Bradshaw Model dictates, as stated in the analysis of the previous hypothesis, that the average depth and width of a river channel increases downstream to the erosional processes present in the river and its growing prominence as one approaches the mouth of the river. As shown in the figures above, the Modau river channel's width has a positive rate of change of 0.2298, as does the channel's depth with 0.0045. This shows that river depth and width follow the predicted characteristics of the Bradshaw Model. It is to be noted, however, that there is a greater increase in the width of the river channel than its depth.

There are anomalies present for the width at Site 6 and 7 as it drops from approximately 7 meters at Site 5 to 5.613m at Site 6 and then further down to 3.873m at Site 7. For the average depth, there is a very stark anomaly present at Site 3, with a depth of 0.417m, where it suddenly shoots up from 0.179m at Site 2, but then afterwards drops down abruptly to 0.18m at Site 4.

4.3.2 Discharge

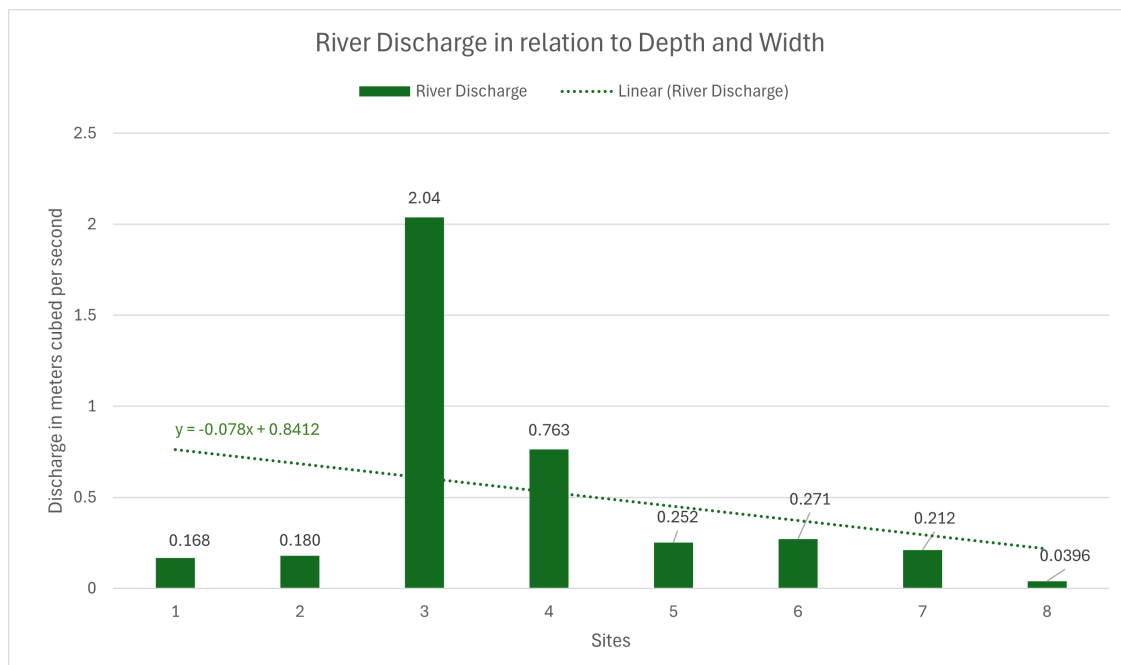


Figure 23: Graph of River Discharge.

In accordance with the Bradshaw Model, the river discharge is said to increase as channel efficiency increases. In the Modau River, there is an increase seen from Site 1, 0.168 cumecs, to Site 3, 2.04 cumecs, however, after then there is a steady decline till Site 8. This does not conform to the predictions of the Model.

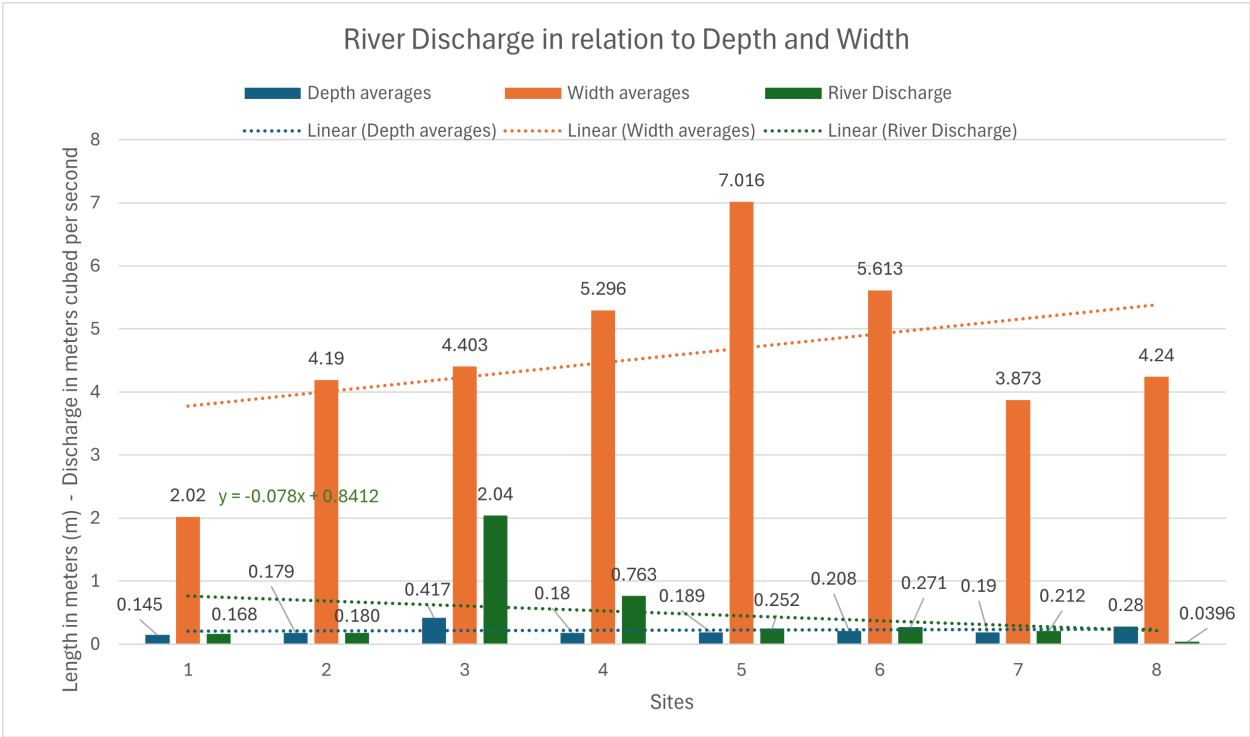


Figure 24: Graph of River Discharge in relation to Depth and Width of the channel at each site.

Shown above is the graph of the River Discharge in relation to the average channel depth and channel width of the Modau River. As portrayed in the graph above, there is no proper correlation in the graph above, as despite the increasing trend of the river’s average depth and width the river discharge has an overall decrease as seen by the line of best fit, with a negative rate of change of 0.078 as shown in Figure 25. Nevertheless, one can see that the river discharge has a moderate to high correlation of 0.595, with the width of the river channel, which is better seen in Figure 25, compared to the its weak, but positive, correlation of 0.333 with the average channel depth.

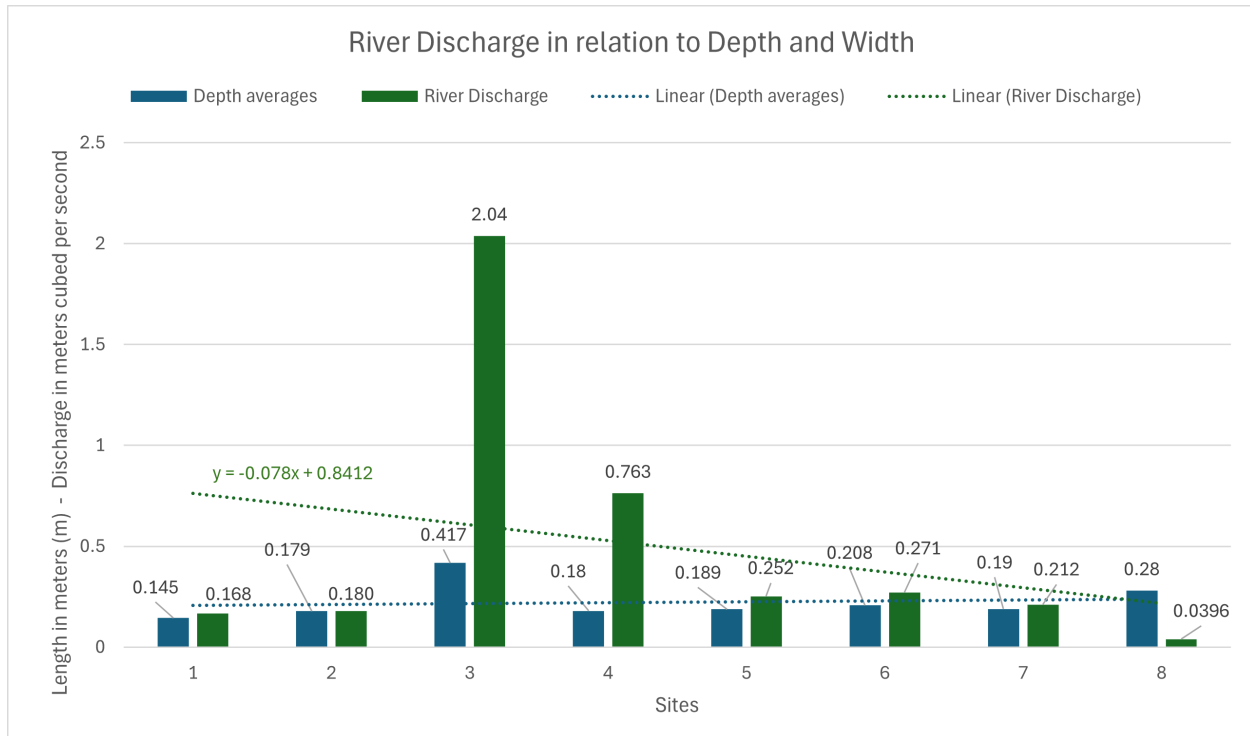


Figure 25: A closer look at River Discharge, but mainly in relation to Depth.

One of reasons for the negative trend of channel efficiency is due to the velocity having an overall decrease as well due to many hard- and soft-engineering techniques. Channel efficiency has an overall decrease due to placements of rip-rap, meanders, and more throughout the middle course and natural sedimentation at the mouth of the river. Nevertheless, this shows us that the data sets don't fully follow the predictions of the Bradshaw Model, and has only a weak to moderate correlation with the depth and width of the river channel. It partially supports the hypothesis as both channel width and channel depth conform to the Bradshaw model, however channel efficiency doesn't.

4.4 Hypothesis 3



Figure 26: Bedload Size at Different Sites

In accordance with the Bradshaw model, bedload size has to decrease as one moves downstream due to the increased amount and impact of erosional processes, especially attrition, corrosion, corrasion, and traction. In the data map above, one can see the clear decreasing trend of the bedload size. There are two anomalies present at Site 6, where the bedload size increases from 38.3 mm at Site 5 to 43.9 mm at Site 6. This is however, justified, as due to a fork in the river at Site 6, with little branches and stones caught at the small nick-point, the flow of the river was slowed down in comparison to Site 5, which had a straight channel with matressing which would instead increase the velocity, causing the erosional processes occurring to be more influential in breaking down the size of the bedload.

This data-set completely supports the hypothesis that bedload size decreases by increased distance away from the source, which in turn also supports the Bradshaw Model.

5 Conclusion

5.1 Hypothesis 1 - Partially supported

The obtained results vary completely from the expected values as although velocity is supposed to increase as channel efficiency increases, the results show the very opposite happening at all sites except for at Site 8, where the data corresponds to one another, yet still doesn't conform to the Bradshaw model, which states that the velocity is highest at the mouth. Channel efficiency does however support the Bradshaw Model as an overall increase is shown as one moves downstream.

5.2 Hypothesis 2 - Partially supported

As mentioned in the hypothesis, increase in the river's discharge does correlate to the channel width and channel depth to a certain extent. As mentioned before there is a moderate correlation between the width and the discharge with an r value of 0.595, however there is a weak correlation between depth and discharge with an r value of 0.333. Channel width corresponds to Bradshaw's Model where it increases as we move downstream as does channel. However discharge decreases as it moves downstream and therefore doesn't support the hypothesis or follow the Bradshaw Model.

5.3 Hypothesis 3 - Fully supported

Results obtained for the Bedload size completely support the hypothesis that they decrease in size as we move downstream, except for at Site 6, where it increases to 43.9mm from 38.3mm before dropping to 23.0mm at Site 7. More importantly, these results wholly support the Bradshaw Model.

6 Evaluation

The strengths of this investigation was that measurements were taken by a specific group of people which allowed for uniformity in the results as the same people measured a specific data set. Pragmatic and Systematic sampling techniques were used, which allowed for more accurate results. Sites that were easily accessible were also chosen, to allow for a more accurate use of fieldwork techniques.

On the the other hand there were also many weaknesses, as some of the equipment used were not ideal for certain measurements, such as the tape measure which was used to measure the wetted perimeter. Furthermore, purposive sampling was also utilised and areas covered with shrubs, algae or large rocks were avoided, forming a biased data set. Furthermore, the time of the year and the time span within which the data was gathered was also highly influential as the water table during summer is lower compared to winter times, and gathering this data on a sunny day yields results different to rainy days. All of these combined form moderate to high biased data sets with room for error.

The Bradshaw model is not ideal for European rivers because it is too simple and assumes everything changes smoothly from the source to the mouth. In reality, European rivers are heavily changed by humans for example, they have dams, weirs, and controlled channels. They are also affected by things like snowmelt, changing seasons, and different land types across the basin. These real-world factors mean European rivers do not follow the neat patterns shown in the Bradshaw model, which makes the model unrealistic for this region (Seibert and Bergström).

A better model for European rivers is the HBV model. This model works well because it includes important processes like soil moisture, snow accumulation and melt, groundwater flow, and rainfallrunoff responses. It can also handle differences across the river basin. Studies show that HBV performs very well in large European rivers such as the Rhine. In fact, research found HBV to be more accurate than another advanced model when tested on

the Rhine basin (te Linde et al.). This makes HBV a much more suitable choice than the Bradshaw model.

7 Appendix

7.1 Raw Data

Sites	Recording 1	Recording 2	Recording 3	Recording 4
Site 1	0.59	0.53	0.60	0.57
Site 2	0.26	0.23	0.23	0.24
Site 3	0.93	1.16	1.24	1.11
Site 4	0.78	0.72	0.90	0.80
Site 5	0.12	0.24	0.21	0.19
Site 6	0.34	0.20	0.16	0.23
Site 7	0.28	0.18	0.40	0.29
Site 8	0.01	0.06	0.03	N/A

Table 1: Velocity Digital

Sites	Cross-sectional Area A (m ²)	Wetted Perimeter P (m)	Channel Efficiency $R = A/P$ (m)
Site 1	0.2929	2.03	0.14428571
Site 2	0.75001	2.96	0.25338176
Site 3	1.836051	6.91	0.26570926
Site 4	0.95328	5.02	0.18989641
Site 5	1.326024	6.75	0.196448
Site 6	1.167504	6.33	0.18443981
Site 7	0.73587	3.53	0.20846176
Site 8	1.1872	4.67	0.25421842

Table 2: Channel Efficiency Calculations per Site

Sites	Recording 1	Recording 2	Recording 3	Recording 4	Recording 5
Site 1	2.01	1.90	2.10	1.98	2.15
Site 2	3.00	3.02	2.95	2.80	3.05
Site 3	6.74	7.00	7.00	6.95	6.85
Site 4	5.12	5.00	4.95	5.09	4.94
Site 5	6.70	6.60	6.85	6.80	6.82
Site 6	6.48	6.20	6.29	6.36	6.30
Site 7	3.60	3.40	3.53	3.57	3.55
Site 8	4.70	4.62	4.78	4.60	4.65

Table 3: Wetted Perimeter

Sites	Recording 1	Recording 2	Recording 3
Site 1	2.00	2.37	1.69
Site 2	2.55	5.98	4.04
Site 3	4.40	4.39	4.42
Site 4	5.28	5.30	5.31
Site 5	7.01	7.01	7.03
Site 6	5.60	5.65	5.59
Site 7	3.83	3.90	3.98
Site 8	4.25	4.24	4.23

Table 4: Width

Sites	Recording 1	Recording 2	Recording 3
Site 1	0.141	0.138	0.155
Site 2	0.312	0.1375	0.084
Site 3	0.426	0.412	0.414
Site 4	0.188	0.182	0.170
Site 5	0.189	0.187	0.190
Site 6	0.213	0.211	0.201
Site 7	0.208	0.158	0.203
Site 8	0.273	0.284	0.283

Table 5: Depth

Sites	Cross-sectional Area A (m ²)	Velocity $\bar{v}$ (m/s)	Discharge $Q = A \times \bar{v}$ (m ³ /s)
Site 1	0.2929	0.5725	0.16768525
Site 2	0.75001	0.24	0.1800024
Site 3	1.836051	1.110	2.03801661
Site 4	0.95328	0.80	0.762624
Site 5	1.326024	0.19	0.25194456
Site 6	1.167504	0.2325	0.27144468
Site 7	0.73587	0.2875	0.21156263
Site 8	1.1872	0.03334	0.03958125

Table 6: Discharge Calculations per Site

7.2 Cross-Sectional Area

Cross-sectional area is measured by multiplying the mean width with the mean depth.

$$A = \text{Mean Width} \times \text{Mean Depth}$$

$$A_{\text{Site 1}} = 2.02 \times 0.145 = 0.2929 \text{ m}^2$$

7.3 Channel Efficiency

1. Channel efficiency = Hydraulic radius; The higher the radius, the greater the efficiency.
2. Cross-Sectional Area $\div$ Wetted Perimeter = Hydraulic Radius

$$R = \frac{A}{P}$$

$$R_{\text{Site 1}} = \frac{0.2929}{2.03} = 0.144 \text{ m (3 s.f.)}$$

7.4 River Discharge

$$\text{Discharge (cumecs)} = \text{Cross-Sectional Area (m}^2\text{)} \times \text{Velocity (m s}^{-1}\text{)}$$

$$Q = A \times v$$

$$Q_{\text{Site 1}} = 0.2929 \times 0.5725 = 0.168(3s.f.) \text{ m}^3/\text{s}$$

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